Hengrong Du

CONTACT INFORMATION

Fisk University
Department of Math & Computer Science

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RESEARCH INTERESTS

Partial differential equations for AI and fluid dynamics

ACADEMIC POSITIONS

Fisk University, Nashville, TN, USA — August 2025-
Assistant Professor, Department of Mathematics and Computer Science
Mentor: **Dr. Qingxia Li**

University of California, Irvine, Irvine, CA, USA — July 2024-July 2025
Visiting Assistant Professor, Department of Mathematics
Mentor: **Dr. Connor Mooney**

Vanderbilt University, Nashville, TN, USA — Sep. 2021-June 2024
Postdoc Scholar (Research), Department of Mathematics
Mentor: **Dr. Gieri Simonett**

EDUCATION

Purdue University, West Lafayette, IN — Aug. 2016-Aug. 2021
Ph.D., Mathematics
Advisor: **Dr. Changyou Wang**

Beijing Normal University, Beijing, China — Sep. 2013-July 2016
M.S., Computational Mathematics
Advisor: **Dr. Jiequan Li**

South China Normal University, Guangzhou, China — Sep. 2009-July 2013
B.S., Mathematics and Applied Mathematics

PUBLICATIONS

Diffusion and Optimal Transport in Machine Learning

1. Xinyang Liu, Hengrong Du, Wei Deng, Ruqi Zhang. *Optimal Stochastic Trace Estimation in Generative Modeling* . AISTATS 2025.

2. Yikun Bai, Rocio Diaz Martin, Hengrong Du, Ashkan Shahbazi, Soheil Kolouri. *Linear Partial Gromov-Wasserstein Embedding* . ICLR 2025.

3. Yikun Bai, Rocio Diaz Martin, Hengrong Du, Ashkan Shahbazi, Soheil Kolouri. *Partial Gromov-Wasserstein Metric* . ICLR 2025.

4. Haoyang Zheng, Hengrong Du, Qi Feng, Wei Deng, Guang Lin. *Constrained exploration via reflected replica exchange stochastic gradient langevin dynamics*. ICML 2024.

5. Wei Deng, Yu Chen, Nicole Tianjiao Yang, Hengrong Du, Qi Feng, Ricky T. Q. Chen. *Reflected Schrödinger bridge for constrained generative modeling*. UAI 2024.

6. Wei Deng, Yu Chen, Nicole Tianjiao Yang, Hengrong Du, Qi Feng, Ricky T. Q. Chen. *On convergence of approximate Schrödinger bridge with bounded cost*. ICML Workshop on New Frontiers in Learning, Control, and Dynamical Systems (2023).

Complex Fluids

1. Hengrong Du, Gieri Simonett, Yuanzhen Shao. *On a thermodynamically consistent model for magnetoviscoelastic fluids in 3D*. J. Evol. Equ. 24 (2024), no. 9, 1-51.

2. Hengrong Du, Yuanzhen Shao, Gieri Simonett. *Well-posedness for magneto-viscoelastic fluids in 3D*. Nonlinear Anal. Real World Appl. 69 (2023), no. 103759.

3. Hengrong Du, Tao Huang, Changyou Wang. *Weak compactness property of simplified nematic liquid crystal flows in dimension two*. Math. Z. 302 (2022), no. 2, 2111-2130.

4. Hengrong Du, Changyou Wang. *Global weak solutions to the stochastic Ericksen–Leslie system in dimension two.* Discrete Contin. Dyn. Syst. 42 (2022), no. 5, 2175-2197.

5. Hengrong Du, Changyou Wang. *Partial regularity of a nematic liquid crystal model with kinematic transport effects.* Nonlinearity 34 (2021), no. 5, 3001-3045.

6. Hengrong Du, Yimei Li, Changyou Wang. *Weak solutions of non-isothermal nematic liquid crystal flow in dimension three.* J. Elliptic Parabol. Equ. 6 (2020), no. 1, 71-98.

7. Hengrong Du, Xianpeng Hu, Changyou Wang. *Suitable Weak Solutions for the Co-rotational Beris–Edwards System in Dimension Three.* Arch. Ration. Mech. Anal. 238 (2020), no. 2, 749-803.

Calculus of Variations

1. Hengrong Du, Nung Kwan Yip. *Stability of self-similar solutions to geometric flows.* Interfaces Free Bound. 25 (2023), no. 2, 155–191.

2. Hengrong Du, Qinfeng Li, Changyou Wang. *Compactness of M-uniform domains and optimal thermal insulation problems.* Adv. Calc. Var. 16 (2023), no. 1, 17-43.

Preprints

1. Hengrong Du, Qi Feng, Changwei Tu, Xiaoyu Wang, Lingjiong Zhu. *Non-Reversible Langevin Algorithms for Constrained Sampling* . arXiv:2501.11743.
2. Yikun Bai, Huy Tran, Hengrong Du, Xinran Liu, Soheil Kolouri. *Fused Partial Gromov-Wasserstein for Structured Objects*. arXiv:2401.03662.
3. Hengrong Du, Chuntian Wang. *Partial regularity for the three-dimensional stochastic Ericksen–Leslie Equations*. arXiv:2401.03662.

Awards

- Bilsland Dissertation Fellowship June 2020-May 2021
  It was intended to give the most accomplished final-year PhD candidates an opportunity to complete the dissertation within the 2020–21 academic year by devoting full-time effort to research and writing.

Talks

- PDE & Applied Mathematics seminar
  University of California, Riverside January 2025
- PDEs & Applications Seminar
  Queen's University January 2025
- Applied Math Seminar
  The University of Alabama October 2024
- PDEs and Fluid Mechanics and Atmospheric Sciences
  West Virginia University April 2024
- AMS Special Session on Fluids: Analysis, Applications, and Beyond
  Florida State University March 2024
- Financial Mathematics Seminars
  Florida State University March 2024
- AMS Special Section on Dynamics and Regularity of PDEs
  Joint Mathematics Meeting January 2024
- Seminar on Analysis and Stochastic Analysis
  Auburn University November 2023
- PDE Seminar
  University of Tennessee, Knoxville October 2023
- PDE/Applied Math Seminar
  Indiana University, Bloomington October 2023
- Richard F. Barry Jr. Seminar Series
  Old Dominion University September 2023
- The 13th AIMS Conference
  University of North Carolina Wilmington June 2023
- AMS Spring Central Sectional Meeting
  University of Cincinnati April 2023
- Undergraduate Seminar in Mathematics
  Vanderbilt University March 2023
- Analysis Seminar
  Wayne State University March 2023

- 3rd Biennial Meeting of SIAM Pacific Northwest Section
  Washington State University — May 2022
- AMS Spring Central Virtual Sectional Meeting
  Purdue University — March 2022
- Graduate Online Mini-course
  Beijing Normal University — November 2021
- PDE Seminar Fall 2021
  Vanderbilt University — September 2021
- Analysis Seminar
  The University of Alabama — February 2021
- PDE Seminar
  Purdue University — September 2020
- Student Analysis Seminar
  Purdue University — February 2020
- Student Analysis Seminar
  Purdue University — February 2020
- PDE Seminar
  Purdue University — January 2020

CONFERENCE PARTICIPATION

- Rivière-Fabes Symposium on Analysis and PDE
  University of Minnesota — Apr. 19-21, 2023
- Shanks Workshop on Advances in Mathematical and Theoretical Biology
  Vanderbilt University — Mar. 17-19, 2023
- Workshop on Geometry and Analysis of Fluid Flows
  Stony Brook University — Jan. 16-20, 2023
- AMS Fall Central Sectional Meeting
  University of Texas at El Paso — Sep. 17-18, 2022
- Shanks Workshop on Mathematical Aspect of Fluid Dynamics
  Vanderbilt University — Feb. 19-20, 2022
- KUMUNU-ISU Conference on PDE, Dynamical Systems, and Applications
  University of Nebraska-Lincoln — Oct. 23-24, 2021
- AMS Fall Western Sectional Meeting
  Online — Oct. 23-24, 2021
- The 84th Midwest PDE Seminar
  Illinois Institute of Technology — Oct. 26-27, 2019
- Midwest Geometry Conference 2019
  Iowa State University — Sept. 6-8, 2019
- The 83rd Midwest PDE Seminar
  Indiana University Bloomington — Mar. 30-31, 2019

TEACHING EXPERIENCE

Instructor at University of California, Irvine — Fall 2024-Present

- MATH 3A Introduction to Linear Algebra — Spring 2025
- MATH 112B Introduction to Partial Differential Equation — Winter 2025
- MATH 3D Elementary Differential Equations — Fall 2024

Instructor at Vanderbilt University — Fall 2021-Present

- MATH 3100 Introduction to Analysis — Spring 2024
- MATH 2400 Differential Equations with Linear Algebra — Fall 2023
- MATH 3100 Introduction to Analysis — Spring 2023
- MATH 3100 Introduction to Analysis — Fall 2022
- MATH 1301 Accelerated Single-Variable Calculus II — Fall 2022
- MATH 2420 Methods of Ordinary Differential Equations — Spring 2022
- MATH 2400 Differential Equations with Linear Algebra — Spring 2022

Recitation Instructor at Purdue University — Fall 2016-Summer 2021

- MA 26200 Linear Algebra And Differential Equations — Fall 2019
- MA 16200 Plane Analytic Geometry And Calculus II — Spring 2019
- MA 16500 Plane Analytic Geometry And Calculus I — Fall 2018